

When Is Dyadic Transformation Reliable?

A Threshold for Relational Event Models with Multi-Receiver Events

Supervisors: Mahdi Shafiee Kamalabad; Myrthe Prins

Introduction

Relational events are everywhere: conversations, messages, meetings – each with a **who**, a **to-whom**, and a **when**.

In this thesis, we look at **Relational Event History (REH)**, a form of data recording these events, with three columns:

Time	Sender	Receiver
10.0	Alice	Bob
12.5	Bob	Alice
18.7	Alice	Carol
22.3	Carol	Alice
27.1	Bob	Carol

Table 1 – REH Data
Each row = one directed interaction, ordered in time.

A **dyadic** event is one interaction between exactly **one sender** and **one receiver** at **one time** — like a direct message to one person, not a group email.

The **Relational Event Model (REM)** (Butts, 2008) is the standard tool for this kind of data, but it requires **one receiver and one unique timestamp per row**.

Many real events have **several receivers at once** (e.g. one email to multiple people). Analysts often **split** each such event into **one row per receiver** so REM can run. On corporate email data such as Enron, about 31% of messages had multiple recipients — and that is considered low for organisational email (Mulder & Hoff, 2024).

Table 2 – Multi-Receiver (MR) Data

One email from Alice to Bob and Carol → two rows, same time.

Time	Sender	Receiver
18.7	Alice	Bob
18.7	Alice	Carol

That split creates **duplicate timestamps** (Table 2) — the problem this thesis quantifies.

Transformation

This thesis applies **tie-breaking timestamp noise**: tiny random delays on duplicate rows only, **limited** so the overall order of events is unchanged.

Pipeline – Timestamp-noise injection

Split multi – receiver event into dyads
→ duplicate timestamps
→ add tiny capped delays on duplicate rows only
→ unique timestamps → fit REM

Research Question

“Up to what proportion of multi-receiver events does separating each multi-receiver event into dyadic events, with a small tie-breaking timestamp noise, preserve REM parameter estimates that remain close to the known dyadic generating parameters?”

Methodology

We simulate event data where five social mechanisms are **known in advance**, then check whether REM still **recovers** them after multi-receiver splitting and tie-breaking noise.

Table 3 – Five effects tracked for parameter recovery

REM Effect	In plain terms
Inertia	Tendency to message the same person again
Reciprocity	Tendency to return a message
OTP	Tendency via a shared contact
Difference (active)	Effect of how “active” sender vs receiver are
Same (group)	More likely within the same group

These are estimated with REM on the transformed data and compared to the simulated truth.

Table 4 – Simulation Design

Setting	Value
Data	Synthetic event histories (Lakdawala et al., 2025; remulate)
Actors / events	20 actors, 2,000 events
Multi-receiver share	0, 2, 8, 16, 24% (+ 12% to locate crossing)
Replication	28 independent baseline datasets
Comparison	Recovered REM vs known truth

When is recovery acceptable?

- At 0% MR: **band** = 2× spread of recovery across 28 datasets (per effect)
- Pass** if bias stays within band → ratio < **1.0** in Figure 1
- Threshold** = highest MR% where all five effects still pass

Pipeline – Full methodology

Simulate dyadic data → inject multi – receiver share → tie – break times → fit REM → compare to truth

Results

Dataset-averaged over **28** simulations: dyadic REM with tie-breaking noise recovers all five effects within tolerance through **8%** multi-receiver share; recovery breaks down by **16%** (**outgoing two-paths first**).

Table 5 – Three zones

Reliable	Borderline	Unreliable
≤ ~8%	~8 – 12%	≥ ~12%

For multi-receiver shares **above ~12%**, prefer a **native multi-receiver model** (e.g. RHEM; Lerner & Lomi, 2023) or another polyadic estimator — not dyadic splitting alone.

Reading Figure 1. Each line tracks one effect from **Table 3**. The y-axis is **|bias| / band** (1.0 = acceptable edge; shaded area = fail), so **rising lines mean worse recovery**. All lines stay below 1.0 through **8% multi-receiver share**. **OTP** crosses first at 16%; **Same (group)** follows. At **12%**, OTP is at **~95%** of its band — the **borderline** zone in Table 5.

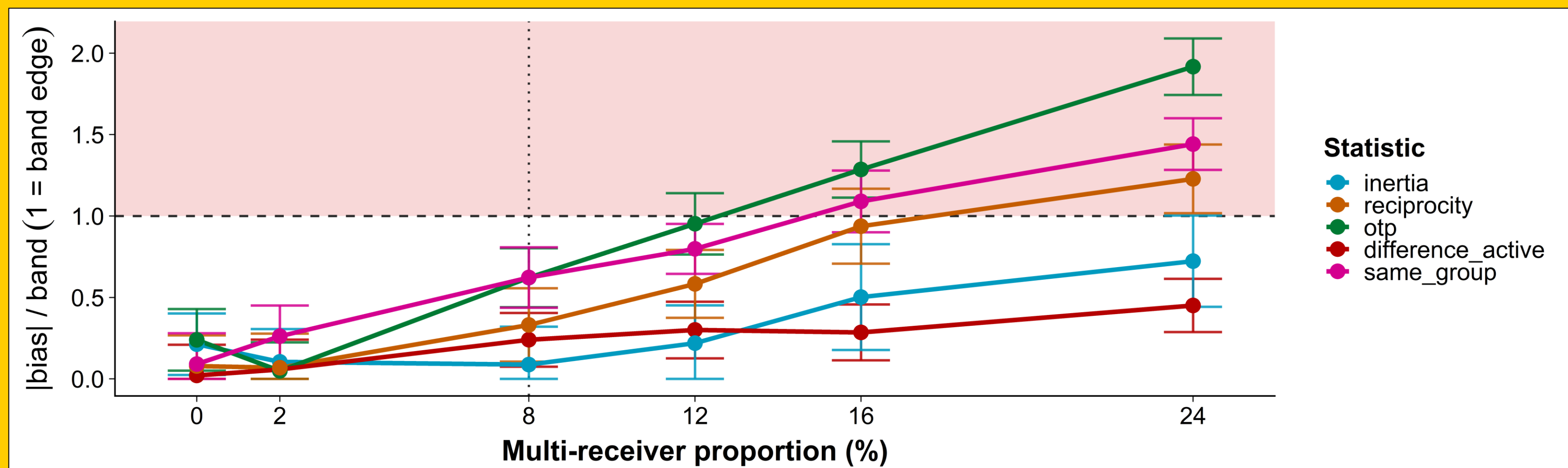
Why this matters

Enron corporate email: **~31%** of messages had multiple recipients — considered low for organisational email (Mulder & Hoff, 2024). A guardrail of **~8–12%** sits **well below** shares common in real communication data.

Scope: Synthetic data only; 20 actors, 2,000 events; random multi-receiver tagging (extra receivers carry no signal); threshold from grid **{0, 2, 8, 16, 24%}** (+ 12% to locate crossing). Structured email lists (fixed CC groups) may behave differently.

Figure 1 – Normalised recovery by multi-receiver share.

y = |bias| / band; 1.0 = edge (shaded = fail). All lines below 1.0 through 8%; OTP crosses first at 16%.



References

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- Lakdawala, R., Mulder, J., & Leenders, R. (2025). Simulating relational event history data: Why and how. *Journal of Computational Social Science*, 8, 92.

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- Lerner, J., & Lomi, A. (2023). Relational hyperevent models for polyadic interaction networks. *Journal of the Royal Statistical Society: Series A*, 186(3), 577–600.
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